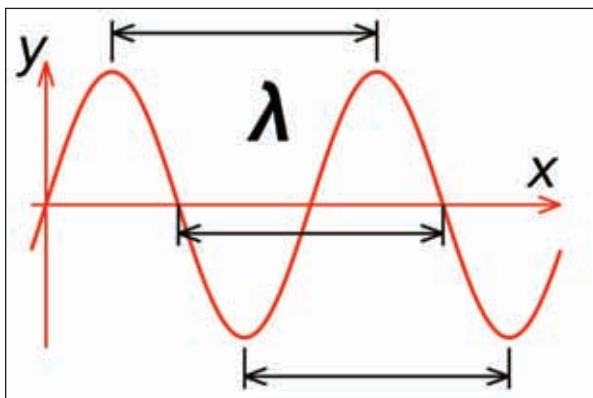




The wavelength is denoted by the Greek symbol lambda

Wavelength

This is the length of a single wave in space as the electromagnetic wave propagates. Mathematically, it is the speed of the electromagnetic wave divided by its frequency.

Speed of the electromagnetic wave

Irrespective of frequency, amplitude and phase, the speed of electromagnetic waves is about 300 million metres per second. This is usually known as the speed of light, light being electromagnetic radiation of much shorter wavelength than radio. It was the discovery that the speed of radio waves and light were the same that led to the conclusion that light is an electromagnetic wave. Visible light, in fact, is an electromagnetic wave in a narrow range of frequencies around about a million gigahertz (10^{15} hertz). Different colours correspond to different frequencies, with red being the lowest and violet the highest. The frequency immediately below the visible range is called infrared and the frequency immediately above is called ultraviolet.

Time period

The time period of a signal is the time taken for one complete variation of the parameter. Mathematically, it is the inverse of frequency and is measured in seconds, milliseconds and microseconds.